Harshit Chaudhary

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OBJECTIVE

Highly motivated **B.Tech (AI/ML)** student seeking a **Quantitative Developer** role to apply strong foundations in **Algorithms, Data Structures, and High-Performance Computing** towards designing and optimizing systematic trading strategies. Eager to leverage practical experience in **MQL5 algorithmic trading** and **time-series analysis** to contribute to production-grade financial systems .

QUANTITATIVE & ALGORITHMIC EXPERIENCE

Algorithmic Trading Development
Mean Reversion Strategy:- MQL5, C++

- **Algorithm Design & Implementation:** Developed the core signal logic combining **Bollinger Bands** (calculated via custom **Standard Deviation** on price data) and **RSI thresholds** to identify oversold/overbought conditions.
- **Performance Engineering:** Implemented **Bar-by-Bar synchronization** (`IsNewBar()` check) and **Time-Based Filtering** to isolate optimal high-liquidity trading hours and enhance execution performance.
- **Numerical Methods:** Performed manual calculation of **Standard Deviation** and **Moving Average** over a price array, showcasing proficiency in fundamental numerical methods essential for financial modeling.
- Improved **Sharp Ratio** and reduce the **Drawdown**.

AlphaSwing Engine
Systematic Trading Engine:- Python

- **Quantitative Scoring:** Developed a **Pandas-based scoring model** using **rank-based percentile analysis** across market factors for final trade signal generation.
- **High-Speed Data Ingestion:** Engineered a **Python ETL pipeline** to handle **real-time JSON data acquisition** from external financial APIs.
- **Production Automation:** Implemented **API-driven notification system** (Telegram) for automated, real-time portfolio updates, showcasing system integration skills.

EDUCATION

Amity University, Gurgaon — July 2023 - July, 2027

Bachelor of Technology in Artificial Intelligence and Machine Learning

- **Quantitative Focus:** Strong foundation in **Data Structures & Algorithms (DSA)**, Numerical Methods, and Time-Series Analysis.
- **Relevant Coursework:** Deep Learning, Machine Learning, **Python Programming**, C++ (implied through DSA), and Big Data Analytics.

SKILLS

- **Programming & Scripting:** Python, C++, SQL, MATLAB, Bash, MQL5
- **Frameworks & Tools:** Flask, MetaEditor, FastAPI, MLflow, DVC, Docker, GitHub Actions, CI/CD